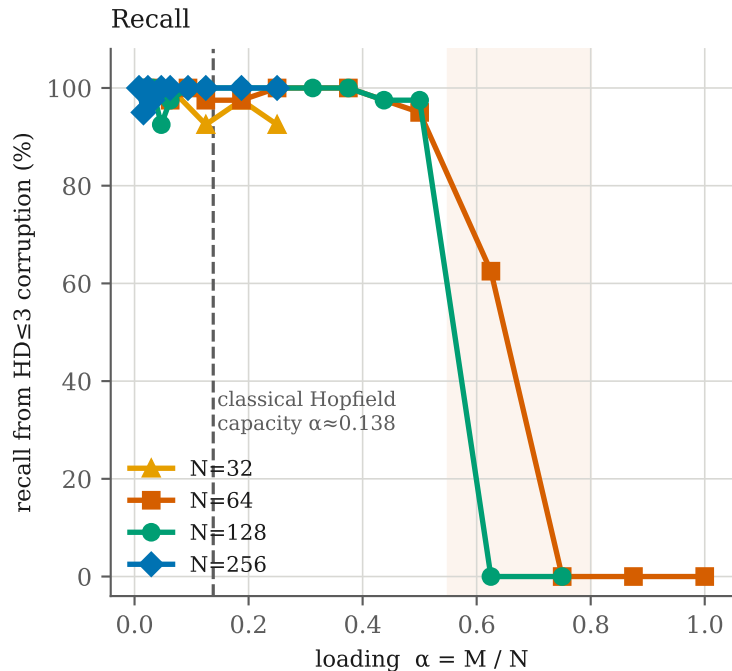
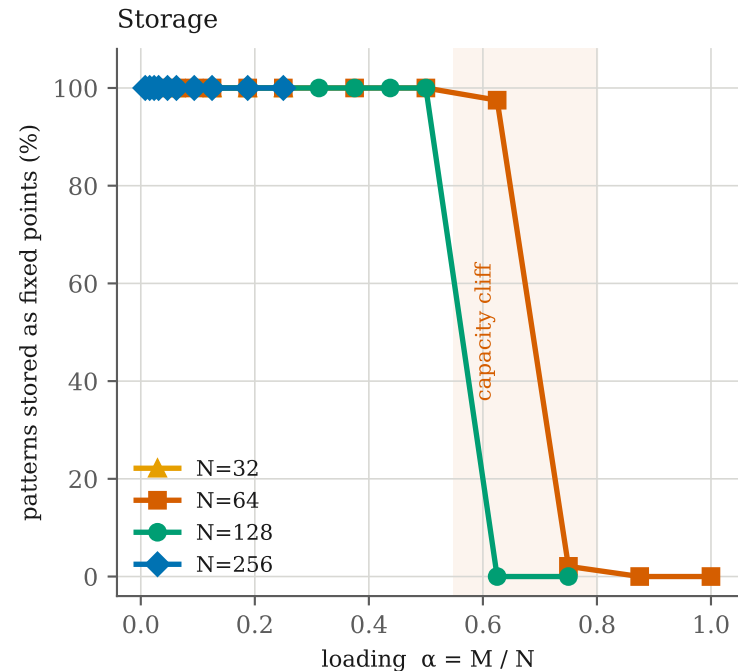


Margin-trained clockless HNN: perfect storage and recall to $\alpha \approx 0.5$, then a sharp cliff



Fan-in grows with M ($\min(N-1, 4M)$), so the high- α points use near-full connectivity — good for the network, expensive for a LUT implementation.